Joseph Poremba

Ph.D. Student in Computer Science, University of British Columbia, Canada

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Education

2022 – 2027 **Ph.D., Computer Science (expected Aug. 2027)**, University of British Columbia
Supervisor: Prof. F. Bruce Shepherd
Thesis: *Models of Multicommodity Network Flow in Series-Parallel and Planar Networks*
Relevant courses: Submodular Optimization, Combinatorial Optimization, Computational Geometry, Machine Learning (audited)

2020 – 2022 **M.Sc., Computer Science**, University of British Columbia
Supervisor: Prof. F. Bruce Shepherd
Thesis: *Directed Multicommodity Flows: Cut-Sufficiency and Forbidden Relevant Minors*
Relevant courses: Program Verification, Formal Methods, Algebraic Methods, Optimization Theory, Distributed Systems Abstractions, Randomized Algorithms

2015 – 2020 **B.Math., Computer Science and Combinatorics & Optimization**, University of Waterloo
Program: Double Major, Honours, Co-operative Education
Supervisor: Prof. Nancy Day
Thesis: *Symmetry Breaking in Many-Sorted Finite Model Finding*
Relevant courses: Algorithm Design and Analysis, Graph Theory, Programming Languages, Groups and Rings, Game Theory, Network Flow Theory

Publications

Publications with N. A. Day have authors ordered by contribution; all others list authors alphabetically by last name, following the convention in theoretical computer science.

- **J. Poremba** and F. B. Shepherd, "A Family of Well-Behaved Multiflow Instances," in *Symposium on Simplicity in Algorithms (SOSA 2027)*. Accepted Oct. 2026, to appear. arXiv: 2611.03858
- C. Chekuri, G. Naves, **J. Poremba**, and F. B. Shepherd, "Uncrossed Multiflows and Applications to Disjoint Paths," in *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques (APPROX/RANDOM 2026)*, LIPIcs, vol. 392, pp. 12:1–12:23, 2026. DOI: 10.4230/LIPIcs.APPROX/RANDOM.2026.12, arXiv: 2511.00254
- R. Dancy, N. A. Day, O. Zila, K. Tariq, and **J. Poremba**, "Portus: Linking Alloy with SMT-Based Finite Model Finding," *IEEE Transactions on Software Engineering*, vol. 52, no. 2, pp. 488–508, Feb. 2026. DOI: 10.1109/TSE.2025.3631361
- D. Alemán Espinosa, N. Kumar, **J. Poremba**, and F. B. Shepherd, "Unsplittable Flow Cut Gap in Undirected Graphs," in *ACM-SIAM Symposium on Discrete Algorithms (SODA 2026)*, pp. 1570–1605, 2026. DOI: 10.1137/1.9781611978971.57
- **J. Poremba** and F. B. Shepherd, "Cut-Sufficient Directed 2-Commodity Multiflow Topologies," *Mathematical Programming*, vol. 210, no. 1–2, pp. 793–814, Mar. 2025. DOI: 10.1007/s10107-025-02195-3 Preliminary version in *Integer Programming and Combinatorial Optimization (IPCO 2023)*, LNCS, vol. 13904, pp. 408–422, 2023. DOI: 10.1007/978-3-031-32726-1_29
- **J. Poremba**, N. A. Day, and A. Vakili, "New Techniques for Static Symmetry Breaking in Many-Sorted Finite Model Finding," *IEEE Transactions on Software Engineering*, vol. 49, no. 6, pp. 3487–3503, Jun. 2023. DOI: 10.1109/TSE.2023.3256939

Presentations

2027 — **Talk (upcoming)**, Symposium on Simplicity in Algorithms (SOSA)
"A Family of Well-Behaved Multiflow Instances"

2026 — **Invited talk (upcoming)**, Workshop on Graphs and Geometric Algorithms (WOGGA 6)
"Uncrossed Multiflows and String Graphs"

2026 — **Talk**, International Conference on Approximation Algorithms for Combinatorial Optimization Problems (APPROX)
"Uncrossed Multiflows and Applications to Disjoint Paths"

2025 — **Invited talk**, Workshop on Integer Programming, Bellairs Research Institute
"Uncrossed Multiflows and Applications"

2024 — **Talk**, International Symposium on Mathematical Programming (ISMP)
"Uncrossable Multicommodity Flows"

2023 — **Talk**, Conference on Integer Programming and Combinatorial Optimization (IPCO)
"Cut-Sufficient Directed 2-Commodity Multiflow Topologies"

2019 — **Poster**, Waterloo Mathematics Undergraduate Research Conference
"Finite Model Finding: Fortress and Symmetry"

Awards and Scholarships

2025 — **CS Graduate TA Award**, University of British Columbia

2022 – 2023 — **BC Graduate Scholarship**, University of British Columbia

2020 – 2021 — **NSERC Canada Graduate Scholarship – Master's**, University of British Columbia

2018, 2019 — **NSERC Undergraduate Student Research Award**, University of Waterloo

2018 — **Electrohome 75th Anniversary Scholarship**, University of Waterloo

Research Experience

2021 – present — **Graduate Research Assistant**, University of British Columbia
Supervisor: Prof. F. Bruce Shepherd
- Researching algorithms and sufficient conditions for multicommodity flow routing, with emphasis on flow-cut gaps (the relationship between the sparsest cut and maximum concurrent flow), and on integral and unsplittable routings

2018, 2019 — **Undergraduate Research Assistant**, University of Waterloo
Supervisor: Prof. Luke Postle
- Researched probabilistic graph colouring, specifically bounds related to Reed's conjecture and strong edge colourings

2017 – 2019 — **Undergraduate Research Assistant**, University of Waterloo
Supervisor: Prof. Nancy Day
- Re-engineered the Fortress finite model finder to improve modularity
- Designed and implemented new symmetry-breaking techniques
- Provided research guidance for new part-time undergraduate research assistants
- Linked George proof checker to Z3 SMT solver for arithmetic proofs

Teaching and Mentorship

Teaching Certificates, University of British Columbia
- Foundations of Pedagogy (2025): term-long course on evidence-based teaching (learning objectives, active learning, assessment, inclusive teaching)
- CS Teaching Development Program (2024–2025): year-long program for Computer Science instructors and graduate students; monthly discussions of research on how students learn and a peer teaching observation
- Instructional Skills Workshop (2023): 24-hour intensive workshop; taught three mini-lessons with peer feedback; earned CIRTL Associate status

2026 – present **Research Mentor, Research Experience (REX) Program**, University of British Columbia
- Mentoring two undergraduate students through a year-long introductory research project on autoformalization: using LLMs to translate human-written proofs into machine-verified proofs in Lean, and studying which kinds of proofs are amenable to it

2024, 2026 **Undergraduate Research Co-mentor**, University of British Columbia
- With Prof. F. Bruce Shepherd, co-supervised two directed-studies students working on confluent flows (2024) and an NSERC USRA undergraduate research assistant working on directed two-commodity flows (2026): assigned exercises, suggested research directions, and reviewed students' work

2021 – 2026 **Graduate Teaching Assistant**, University of British Columbia
- CPSC 420, advanced undergraduate algorithms (2022–2026): taught lectures on multiplicative weights, graph colouring, and Max-SAT; designed homework problems; ran review sessions
- CPSC 536S, graduate combinatorial and submodular optimization (2025, 2026)
- CPSC 320, intermediate undergraduate algorithms (2024)
- CPSC 121, models of computation (2021): ran tutorials and review sessions

2018 **Undergraduate Teaching Assistant, MATH 135/136**, University of Waterloo
- Assisted students with algebra courses in the Math Tutorial Centre

Service

2022 – present **Reviewer**
Conferences: FOCS 2022; ESA 2022; SODA 2022, 2027; IPCO 2025; SOSA 2026
Journals: SIAM Journal on Computing (2024); Discrete Optimization (2025)

Conferences & Workshops Attended

2027 SOSA (upcoming)

2026 WOGGA (upcoming); APPROX; SODA

2025 Bellairs Workshop on Integer Programming

2024 STOC; SODA; ISMP; Fulkerson 100 Workshop; Graduate Students in Teaching Conference

2023 IPCO; IPCO Summer School

Selected Course Projects

2022 **Greedy Routing Evaluation**, CPSC 516
- Implemented and empirically evaluated a Schnyder-wood-based greedy routing algorithm for planar graphs; in Python

2021 **Program Verification**, CPSC 539S
- Formally verified algorithms in the Viper verification language
- Implemented a program verifier for a Viper-based imperative language, generating correctness formulas and interfacing with an SMT solver; in Scala

2021 **Distributed Graph Processing System**, CPSC 538B
- Designed and implemented a system for distributed execution of graph algorithms such as shortest paths, PageRank, and graph colouring; in Scala

Other Work Experience

2017 **Software Developer**, Gemalto (now Thales), Ottawa, ON
- Developed a JSON Web Token authentication plugin for a C++ REST API
- Led a one-month self-directed project demonstrating the feasibility of a potential new service, culminating in a technical presentation and demo

2016 **Software Developer/Tester**, DF/Net Software ULC, Dundas, ON
- Developed an XLSX data export library in C++, C, and Qt with custom styling support; profiled it with Callgrind and cut the runtime of large exports by 80%
- Investigated password hashing protocols to integrate into company solutions